Preparation memory in the complete bilocal covariance

A stationary spectral term and a neutral Gram correction

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Abstract

The single-insertion projection does not preserve the full bilocal fluctuations. Nevertheless the change from the energy-tilted state to the actual single-selected state has a simple two-insertion form: its uncentered part is a positive Mehler Gram kernel built from the projected neutral tower. All charged, neutral and mixed residual fluctuations remain in an independently specified stationary spectral term. The connected preparation correction includes subtraction of the nonzero means and need not be positive. We prove its decay under a common large center-time translation after double scaling, and compute its leading t^6 and t^7 behavior.

1 Scope, units and state

Retain the overlapping complex Gaussian p-body CAR model with the same original disorder, half filling, and $p^2/N \to \lambda > 0$. Exact finite-N pinching precedes the double-scaling limit. Set

$$C = J_s^2/\lambda, \quad q = e^{-4\lambda}, \quad \zeta = \sqrt{q}, \quad h = H/\sqrt{C}, \quad W = \sqrt{\lambda}U, \quad \hbar = 1.$$

The orthogonal fields h, U have unit q-Gaussian variance. Our times are $t = \sqrt{C}\, t_{\mathrm{phys}}$ and $T = \sqrt{C}\, T_{\mathrm{phys}}$, not the string variable $u = 2\sqrt{\lambda}T$. The two normalized positive states are

$$\sigma(Z) = \tau(\rho Z), \qquad \rho = 1 - \zeta(1+\lambda) + \zeta(h-W)^2, \tag{1}$$

$$\tau_f(Z) = \tau(f(h)Z), \qquad f(h) = 1 - \zeta + \zeta h^2. \tag{2}$$

The first is the actual single-selected preparation identified in [4, 7]; the second is its energy-only restriction. They are different states on the full neutral algebra.

Let $\mathcal{A}$ be the closed h, U polynomial subspace in the joint tracial L^2 space, and P its orthogonal projection. The preceding result [9] gives, for every energy-centered single-mark descendant X in the charged/neutral tower,

$$\mathsf{P}X = \sum_{a,b} \beta_{X,ab} h^a W h^b, \qquad \beta_X(x,y) = \sum_{a,b} \beta_{X,ab} x^a y^b. \tag{3}$$

Charged J_d have $\beta_{J_d} = 0$; this does not make J_d zero. Energy dressing, finite sums and the already controlled time limits are included. Write

$$d\nu(x,y) = K_q(x,y) d\mu_q(x) d\mu_q(y), \quad (\mathcal{C}\beta)(x) = \int K_q(x,y)\beta(x,y) d\mu_q(y).$$

The existing q-Mehler kernel is normalized and positive [1]; hence $\|\mathcal{C}\beta\|_2 \le \|\beta\|_{L^2(\nu)}$. The normalization (3) gives $\|\mathsf{P}X\|_2^2 = \lambda \|\beta_X\|_{L^2(\nu)}^2$.

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2 The two-insertion source identity

Theorem 1. *For energy-centered single-mark descendants X, Y, including their h-dressings and the controlled finite-time limits,*

$$\boxed{\sigma(X^\dagger Y) = \tau_f(X^\dagger Y) + \zeta\lambda^2(1+q)\langle \mathcal{C}\beta_X, \mathcal{C}\beta_Y\rangle_{\mu_q}.} \tag{4}$$

The means and the connected covariance are therefore

$$\mu_X := \sigma(X) = -\zeta\lambda \int (x+y)\beta_X(x,y)d\nu(x,y), \tag{5}$$

$$\mathrm{Cov}_\sigma(X,Y) = \tau_f(X^\dagger Y) + \Gamma(X,Y), \tag{6}$$

$$\Gamma(X,Y) = \zeta\lambda^2(1+q)\langle \mathcal{C}\beta_X, \mathcal{C}\beta_Y\rangle - \overline{\mu_X}\mu_Y. \tag{7}$$

No replacement of $X^\dagger Y$ by $(\mathsf{P}X)^\dagger \mathsf{P}Y$ is made in the first term.

Proof. The state difference is $\rho - f = \zeta[(W^2 - \lambda) - \{h, W\}]$. Use the established nilpotent source expansion for disjoint whole-Wick D/L marks [3, 4]. Each centered boundary W has no singleton source. Its only connected sources are: another W at the opposite end of the same chord, with coefficient one, or an L block containing that opposite end, also with coefficient one. Connections to D vanish. A source cannot connect to two others, because all source cumulants of order three or higher vanish in this class.

For two adjacent boundary W's, the pairing of those two slots is an adjacent Gaussian chord. Its contribution is exactly $\lambda\tau(X^\dagger Y)$ and is canceled by $-\lambda$. In every remaining source partition each boundary must attach to a different field mark. Replacing the incident slot in each field by W gives exactly its one-U projection from [9], with the same whole-Wick exclusions, crossing weights and chaos counts. Thus

$$\tau[(W^2 - \lambda)X^\dagger Y] = \tau[(W^2 - \lambda)(\mathsf{P}X)^\dagger \mathsf{P}Y]. \tag{8}$$

Linear combinations and energy-regression subtractions preserve this identity. It includes lower-chaos terms by keeping the original field's outgoing-chord count in each diagram.

For one boundary W in $\{h, W\}$, it must attach to one field. That field is replaced by its one-U term; the remaining field can then be projected against the resulting h, U test word. Its energy regression is zero. The surviving projected word has three U's, so its trace vanishes. Therefore

$$\tau[\{h, W\}X^\dagger Y] = 0. \tag{9}$$

Finally the known adjacent-four-U identity [8] gives

$$\tau[(U^2 - 1)f(h)Ug(h)Uk(h)] = (1+q)\int f(a)g(b)k(c)\, d\Pi(a,b,c),$$

where $d\Pi = d\mu(a)K_q(a,b)d\mu(b)K_q(b,c)d\mu(c)$. Substitution of (3) gives the Gram term in (4); (5) is the preceding one-insertion formula. Energy centering implies $\tau_f(X) = \tau_f(Y) = 0$. □

3 What remains outside the projection

Write $X = P_X + Q_X$, $P_X = \mathsf{P}X$, and similarly for Y. For finite-core words, $\rho P_X^\dagger$ and $f P_X^\dagger$ are valid $\mathcal{A}$ test vectors. Orthogonality therefore kills the cross terms in either state. The formulas extend by the bounded densities and the inherited L^2 limits. Theorem 1 also applies to Q_X, Q_Y by linearity, because they are sums of single-mark terms and their projected polynomials. Their projected symbols vanish. Consequently

$$\mathrm{Cov}_\sigma(X,Y) = \tau_f(Q_X^\dagger Q_Y) + \mathrm{Cov}_\sigma(P_X, P_Y). \tag{10}$$

Both kernels on the right are positive. The first includes the nontrivial charged–neutral and neutral–neutral residual sectors; their norms have not been discarded.

For example, the centered charged curvature obeys

$$\mathsf{P}J_2 = 0, \qquad \sigma(J_2^2) = \tau_f(J_2^2) = 4\lambda + \frac{16\lambda^2 q}{1+q} > 0.$$

Substituting only projected fields would lose this entire leading contribution to the bilocal fluctuations.

The connected excess is not positive in general. The Gram addition in (4) is positive, but the state difference in connected covariance also subtracts means. Take the centered pure neutral size-two field $X = S_2 - \kappa_2 H_2$, whose projected symbol is $\beta_X = x + y$. Since $\mathcal{C}\beta_X = (1+q)x$, one obtains

$$\mu_X = -2\zeta\lambda(1+q), \tag{11}$$

$$\Gamma(X,X) = \zeta\lambda^2(1+q)^2(1+q-4\sqrt{q}). \tag{12}$$

This is strictly negative for $q > (2-\sqrt{3})^2$ on the physical curve. It contradicts neither positivity of the full selected variance nor (10). An independent rational-source check at $q = 16/81$, $\lambda = 1/5$ gives exactly

$$\Gamma(X,X) = -\frac{1768892}{119574225}, \qquad \mathrm{Var}_\sigma(X) = \frac{806826868}{773246655} > 0.$$

This rational point checks the algebra; its parameters are not being claimed to satisfy $q = e^{-4\lambda}$.

A controlled scope. The proof concerns products of two fields, each linear in one marked insertion before taking sums and time limits. It does not identify a conditional expectation of arbitrary products with the product of their projections. Four bilocal insertions have additional internal source partitions and must retain them.

4 The complete bilocal two-point function

The preceding mean calculation provides the entire symbol

$$\beta_{T,t}(x,y) = -2e^{iT(x-y)} \sum_{j\geq 1} \Xi_j(x,y) L_j(t;x,y), \tag{13}$$

$$L_j(t;x,y) = \int_0^t (t-s)\,\mathrm{sinc}\left(\frac{(t-s)(x-y)}{2}\right) e^{-is(x+y)/2} c_j(s)\,ds. \tag{14}$$

Here $\mathsf{P}\mathcal{B}(T,t)$ has W-symbol $\beta_{T,t}$, $c_j(s) = \int e^{isx} H_j(x) d\mu_q(x)/[j]_q!$, and the lower-chaos recursion for Ξ_j is kept exactly as in [9]. The actual double time filter and its sign are already incorporated.

For completeness, the baseline in (6) is specified constructively by the existing full joint spectral measure [5, 7], not by a new independent-probe model. Let X_a range over charged coefficients L_n and neutral coefficients N_j, with

$$k_{L_n}(s) = \frac{(is)^n}{n!}, \qquad k_{N_j}(s) = 2c_j(s).$$

The common matrix measure is oriented as

$$d\nu_{ab}(x,y) = \tau[X_a^\dagger P_h(dx) X_b P_h(dy)],$$

so x is the output energy and y the input energy. Its mixed entries retain both D–L connected sources and the same disorder. Define

$$b_a(T,t;x,y) = -e^{iT(x-y)} \int_0^t (t-s)\,\mathrm{sinc}\left(\frac{(t-s)(x-y)}{2}\right) e^{-is(x+y)/2} k_a(s)\,ds.$$

Then the stationary energy-tilted term is

$$\mathcal{S}_f(1,2) = \sum_{a,b} \int f(y) \overline{b_a(T_1,t_1;x,y)} b_b(T_2,t_2;x,y)\, d\nu_{ab}(x,y). \tag{15}$$

It depends on the centers only through $T_2 - T_1$. All fixed-field moments of this measure are given by the existing joint-source rules, and its absolute-continuity and time-tail construction are inherited from [5, 6, 7].

The result for the full actual bilocal is

$$\boxed{\mathrm{Cov}_\sigma(\mathcal{B}_1,\mathcal{B}_2) = \mathcal{S}_f(1,2) + \zeta\lambda^2(1+q)\langle \mathcal{C}\beta_1, \mathcal{C}\beta_2\rangle - \overline{\mu_1}\mu_2.} \tag{16}$$

The density f weights the *input* energy in (15). No extra factor of C appears: the physical memory's factor C cancels its two physical time differentials. The numerical results below evaluate the preparation terms and common-center differences, not the absolute baseline integral.

5 Convergence and loss of preparation memory

For the projected symbol, let $r = (1-q)^{-1/2}$, $F_q = 4\lambda q/(1-q)$, $A = qr|t|$, $B = 3r|t|$. The preceding proof gives $|c_j(s)| \le |s|^j/j!$ and $\|\Xi_j\|_2 \le 4\lambda j(qr)^j + F_q(3r)^j$. Thus a bound for both full and truncated W-symbols is

$$M(t) = t^2[4\lambda A e^A + F_q(e^B - 1)]. \tag{17}$$

For a degree cutoff J satisfying $J+1 > A$, $J+2 > B$, set

$$d_J(t) = t^2 \left[\frac{4\lambda A^{J+1}}{J!\,[1 - A/(J+1)]} + \frac{F_q B^{J+1}}{(J+1)!\,[1 - B/(J+2)]} \right]. \tag{18}$$

Then $\|\beta - \beta_{\le J}\|_{L^2(\nu)} \le d_J(t)$, uniformly in real T. Put $c = \zeta\lambda^2(1+q)$ and $a = \zeta\lambda\sqrt{2(1+q)}$. The contraction of $\mathcal{C}$ and Cauchy–Schwarz give the following conservative bound for the connected preparation part:

$$|\Gamma(1,2) - \Gamma_{\le J}(1,2)| \le (c + a^2)[d_1 M(t_2) + M(t_1)d_2 + d_1 d_2], \qquad d_i = d_J(t_i). \tag{19}$$

These are exact-arithmetic degree estimates, excluding quadrature and roundoff. The baseline converges by the established full-field tails: if $\|\mathcal{B}_i - \mathcal{B}_i^{(D)}\|_2 \le \delta_i$, then its error is bounded by

$$\|f\|_\infty[\delta_1\|\mathcal{B}_2\|_2 + \|\mathcal{B}_1^{(D)}\|_2\delta_2], \qquad \|f\|_\infty = 1 - \zeta + \frac{4\zeta}{1-q}.$$

The actual sector/preparation bridge [7] supplies the finite-N identification. Limit-only bounds are not used to infer an unproved microscopic rate.

Theorem 2. *After the fixed-λ double-scaling limit, for fixed internal times t_i and fixed center offsets T_i,*

$$\lim_{|R|\to\infty} \Gamma(T_1 + R, t_1; T_2 + R, t_2) = 0. \tag{20}$$

The selected connected two-point function consequently approaches the stationary baseline (15) *under a common translation.*

Proof. A common translation multiplies $\beta_i(x,y)$ by $e^{iR(x-y)}$. For a bounded polynomial symbol the conditional integral in y tends to zero for each x by the Riemann–Lebesgue lemma: $K_q(x,y)d\mu_q(y)$ has an integrable density. Dominated convergence gives $L^2(\mu_q)$ convergence. Polynomial symbols are dense in $L^2(\nu)$, and the conditional map is a contraction for every R, so approximation extends the claim to the complete symbols. The same argument applies to $y\beta_i$ because energy support is compact. Since $\mu_i = -\zeta\lambda\int[x\mathcal{C}\beta_i + \mathcal{C}(y\beta_i)]d\mu_q$, the means also vanish. Equation (7) completes the proof. □

The order is double scaling, then large common center time. No finite-N recurrence statement, rate, or uniform joint limit is asserted.

6 Short times and a full-variance difference

The projected symbol starts as

$$\beta_{T,t} = e^{iT(x-y)}\left[-\frac{4i}{3}\lambda q\, t^3 - \frac{\lambda q(1-q)}{3}(x+y)t^4 + O_{L^2}(t^5)\right]. \tag{21}$$

Define the existing two-energy correlation

$$C_2(T) = \int e^{iT(x-y)} K_{q^2}(x,y)\, d\mu_q(x)\, d\mu_q(y), \qquad D_q = \frac{16}{9}\sqrt{q}\,\lambda^4 q^2(1+q).$$

By the Mehler semigroup, $\|e^{iTx}P_q e^{-iT\cdot}(x)\|_2^2 = C_2(T)$. To retain the next term, use the stationary three-energy chain:

$$\int (x+z)e^{iT(y-z)}\, d\Pi(y,x,z) = \frac{i}{2}C_2'(T).$$

Indeed its real part vanishes by simultaneous sign reversal; subtracting its conjugate exchanges y, z, leaving $\int (z-y)e^{iT(y-z)}d\Pi = iC_2'(T)$. Substitution in (7) yields

$$\Gamma(T,t;T,t) = D_q\left[C_2(T)t^6 + \frac{1-q}{4}C_2'(T)t^7\right] + O(t^8). \tag{22}$$

The mean product first contributes at t^8. The t^7 term is generally nonzero at nonzero center; the variance must not be assumed even in the internal time in this nonstationary state.

For equal internal times the stationary baseline is independent of common center. Therefore the following is a difference of *full* bilocal variances, not merely of projected ones:

$$\begin{aligned}
&\mathrm{Var}_\sigma\, \mathcal{B}(T,t) - \mathrm{Var}_\sigma\, \mathcal{B}(0,t) \\
&\quad = \Gamma(T,t;T,t) - \Gamma(0,t;0,t) \\
&\quad = D_q\left[(C_2(T)-1)t^6 + \frac{1-q}{4}C_2'(T)t^7\right] + O(t^8).
\end{aligned} \tag{23}$$

The q-Hermite spectral expansion gives $0 < C_2(T) < 1$ for $T \neq 0$: it is a positive weighted sum of squared Fourier coefficients, with weights q^{2n}, and equality at one requires e^{iTx} to be constant on the continuous energy support. Thus the leading center-time change is negative.

The full variance itself starts at an earlier order,

$$\mathrm{Var}_\sigma\, \mathcal{B}(T,t) = \frac{t^4}{4}\left(4\lambda + \frac{16\lambda^2 q}{1+q}\right) + O(t^5),$$

because $\mathcal{B}(T,t) = -t^2\alpha_T(J_2)/2 + O_{L^2}(t^3)$ and the single-selected curvature restriction is stationary. This leading term would be lost by multiplying single-field projections. The formulas above concern ensemble/model limits with the actual inherited preparation, not independent probes or finite-G scattering.

7 Executed checks and numerical scope

Exact two-field checks. The original local CAR source evaluator with the true selected boundary was compared against (4) at $(q, \lambda) = (1/16, 1/3), (16/81, 1/5)$. It includes prefix neutral marks, charged J_2, J_3, the actual ordered $N_1, \dots, N_4$ recursion, lower-chaos/OU terms, both field orders and inserted energy powers. The first successful output contains 386 full-selected comparisons, 16 separate odd-boundary zeros, 16 separate even-boundary identities, 2 charged marginal checks and 2 residual Gram matrices. All 14 principal minors of those two 3×3 matrices are nonnegative as exact rationals. A separate exact witness verifies the negative centered excess in Section 3. These formal rational parameters test algebra and do not lie on the physical curve.

Fixed-λ time diagnostics. On the physical curve use $q = 1/4, 1/2$, $\lambda = -\log q/4$, internal times $t = 1/4, 1/2, 1$ and centers $T = 0, 1/2, 1, 2, 4$. We compare $(J, L, n_s) = (24, 48, 24)$ with $(32, 64, 32)$, where L is the energy Jacobi dimension and n_s the time quadrature order. Scalar c_j coefficients are summed starting at their first possible Taylor order j; this avoids high-degree eigensolver subtraction. Ten old mean values are reused as read-only calibration and agree without rerunning the old driver.

Across 30 diagonal cases the maximum absolute change in Γ is $3.26 \cdot 10^{-19}$, and the maximum relative change is $8.34 \cdot 10^{-14}$. The largest analytic degree bound (19) is $3.51 \cdot 10^{-16}$. These bounds exclude energy/time quadrature and floating-point error; refinement agreement is a diagnostic, not an interval certificate. Six short-time cases at $T = 1/2$ check the t^6, t^7 terms and record the residual divided by t^8.

q	t	T	$\operatorname{Var}\mathcal{B}(T,t) - \operatorname{Var}\mathcal{B}(0,t)$
$1/4$	$1/2$	$1/2$	$-3.9458574240 \cdot 10^{-6}$
$1/4$	1	$1/2$	$-2.2970480041 \cdot 10^{-4}$
$1/4$	1	1	$-5.0423075726 \cdot 10^{-4}$
$1/4$	1	2	$-6.7204451036 \cdot 10^{-4}$

The table follows from exact cancellation of the common baseline in (23); absolute total variances were not numerically integrated here. All complex values and refinement differences are retained in the output.

Two 6×6 unequal-time uncentered correction matrices are Hermitian and numerically positive. Their connected excess matrices instead have minimum eigenvalues about $-8.19 \cdot 10^{-7}$ and $-2.21 \cdot 10^{-7}$, consistent with the exact warning in Section 3. No negative eigenvalue is clipped or reinterpreted as a failure of the full covariance's positivity.

8 Next physical comparison

We now have an actual selected mean and a full two-point reduction with the nonstationary preparation terms explicit. The stationary baseline still carries the entire same-disorder charged/neutral spectral matrix. This is a useful point to return to the static-patch dictionary: compare the state, time units, energy spectrum and candidate spatial/mass generators before interpreting any Mehler exponent as a bulk particle mass.

Selected four-bilocal correlators require additional source partitions and remain part of the dynamical program. The current two-insertion identity cannot be iterated by treating products as single-mark fields. Neither it nor its large-center-time limit establishes finite-G gravitational scattering or a complete de Sitter holographic dual.

References

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